

4.4 Entity Relationship Model (ER-Model)

ای آر ماڈل ڈیٹا بیس کا خاکہ ہے۔

CS XI

Chapter 4: Data and Analysis

Lecture 06 of 15

★ Golden

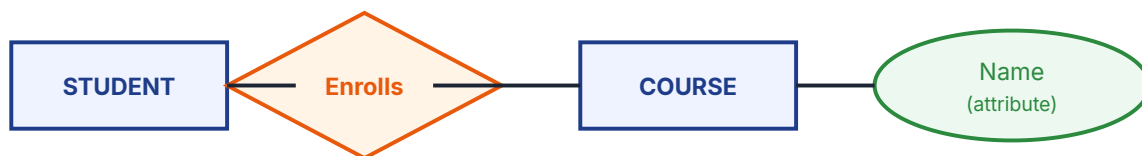
4.4 Entity Relationship Model (ER-Model)

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LEARN IT

The **ER model** is the architect's blueprint: you draw what will be stored and how it connects *before* you create tables in Access. Three building blocks:

Component	Meaning	Symbol	Example
Entity	A real-world object we store data about	Rectangle	STUDENT, TEACHER, COURSE
Attribute	A property of an entity. PK is underlined	Oval	StudentID, Name, Class
Relationship	How two entities are connected	Diamond	Enrolls, Teaches, Has



Rectangle = entity, diamond = relationship, oval = attribute. Underline the primary-key attribute.

Cardinality (types of relationship)

Type	Meaning	Example
One-to-One (1:1)	One record $\leftrightarrow$ exactly one record. Rare; often for security.	One student has one ID card.
One-to-Many (1:M)	One record in A $\leftrightarrow$ many in B. The most common type.	One teacher teaches many students.
Many-to-Many (M:N)	Many in A $\leftrightarrow$ many in B.	Many students enrol in many classes.

COMMON MISTAKE

The M:N rule. A relational database cannot store M:N directly — it duplicates data. Break it into two 1:M relationships with a **junction table** (also called a bridge or associative table). Example: Enrollment(StudentID, ClassID, Date).

EXAM TIP

Always state cardinality on the diagram (1 and M at the ends of the line). For M:N, the long question is “how do you resolve it?” — draw the junction entity with two foreign keys.

اردو میں سمجھیں

ای آر ماڈل ڈیٹا بیس کا خاکہ ہے۔ مستطیل ہستی، بیضوی صفت، لوزی تعلق۔ تعلقات 1:1، 1:M یا M:N ہوتے ہیں۔ M:N کو دو M:1 میں توڑنے کے لیے جنکشن ٹیبل بنانا پڑتا ہے۔

CHECK YOURSELF

1. What symbol is used for a relationship?
2. A hospital: one patient, many appointments. Cardinality?

Answers: 1. Diamond. 2. 1:M (Patient 1 — has — M Appointment).

Next lecture: 4.5 Referential Integrity